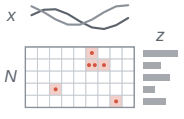
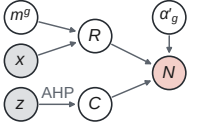
Detection layer



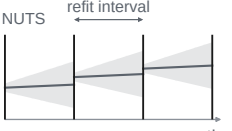
N_{it}, x_{it}, z_i

Hierarchical NB model



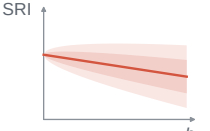
$N \sim \mathrm{NB}(RC\Delta,\ \alpha'_g)$

Posterior computation



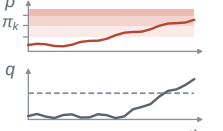
refit and propagate

Uncertainty



posterior predictive

Two-channel alarm



triage p and change q